PROJECT 6

STUDENT PERFORMANCE

Simple Linear Regression with Single Predictor

PRESENTED BY MUNAWAR

BACKGROUND

Modern education now views academic achievement and non-academic involvement as two key pillars in student competency development at school. This study was conducted to evaluate collective achievement standards by calculating average grades to map students' overall academic standing.

The primary objective of this study was to determine the overall parameters of student success by calculating their average scores. Furthermore, an analysis was conducted to identify the strength of the relationship and the influence of extracurricular activities on student academic outcomes using a linear statistical approach.

EXPLORATORY DATA ANALYSIS

4.234	10.55	0.2078	0
Data Sample	Avg. Student Score	Standard Error	Missing Value

- The value 10.55 is also an estimate of the intercept of a null model (without predictors), because $\hat{a} = \bar{y}$ when there are no predictors in the model.

Goal

- Analyze the relationship between extracurricular participation and student scores using Simple Linear Regression.

OLS LINEAR REGRESSION MODEL

Simple regression built in this equation

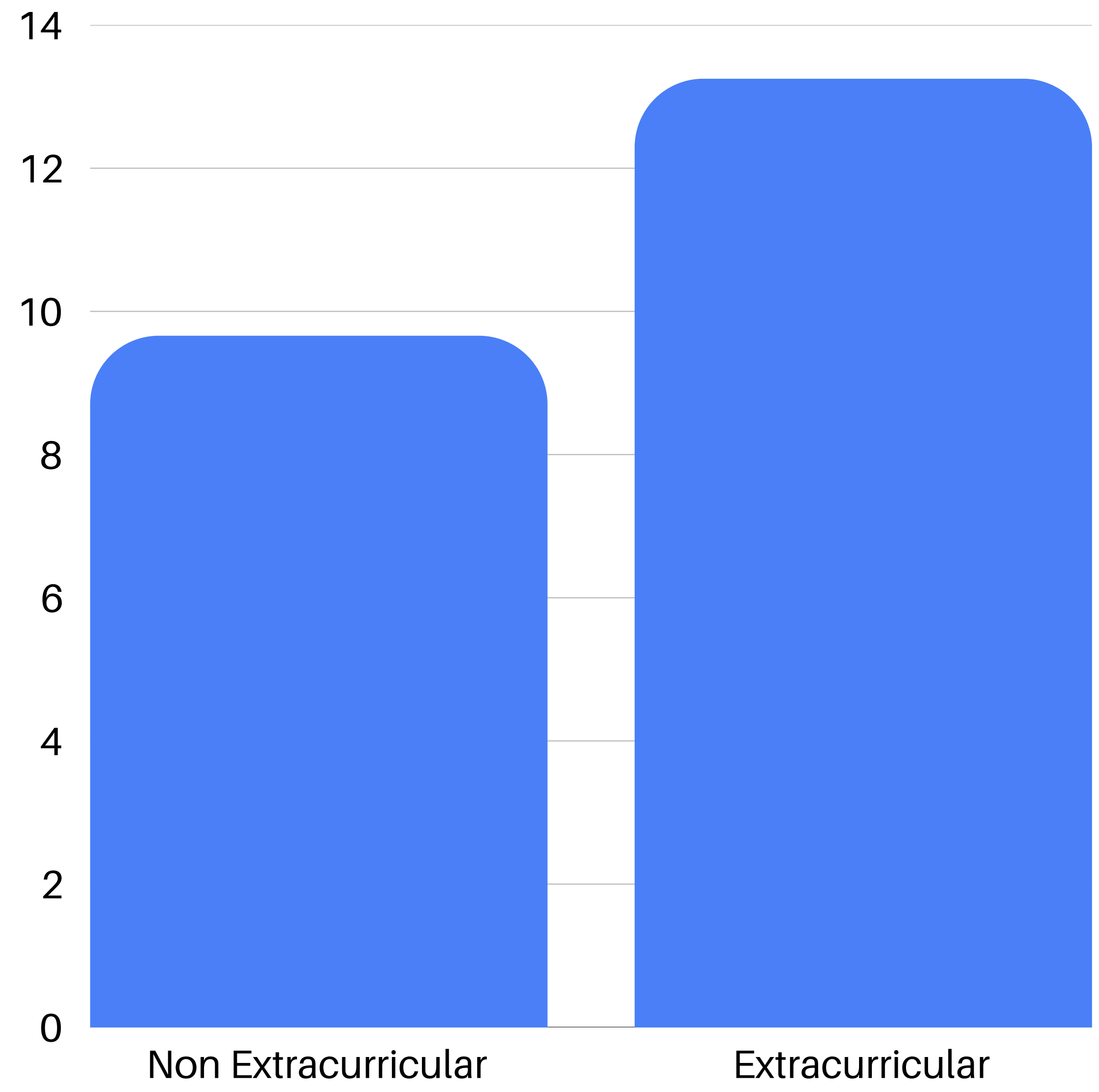
$$y = a + bx + \varepsilon$$

- y : output / outcome
- x : input variables / predictor
- a : intercept
- b : slope

$$y = 9.6692 + 3.5834x$$

Parameter	Estimate	Std Error	Interpretation
Intercept	9.6692	0.2803	Average score of students who do NOT participate in extracurricular activities (extracurricular = 0)
Session Duration	+3.5834	0.4134	Students who participated in extracurricular activities had an average score of 3.58 points HIGHER

COMPARISON OF AVERAGE SCORES PER GROUP



9.66
Non Extracurricular

13.25
Extracurricular

+3.59
Slope

CONCLUSION

1. The average score of all students is **11.31**. This value can be interpreted as the best estimate of a student's score when no predictor information is available.
2. The regression model shows that students who participate in extracurricular activities have an average score of **3.59 points higher** than those who do not participate in extracurricular activities. This difference is mathematically consistent with the mean difference between groups, demonstrating the reliability of the model for binary predictor variables.
3. This analysis only considers one predictor (extracurricular). The dataset contains other variables such as study_hours, attendance, parental_education, and region that could potentially provide a more comprehensive explanation of student scores.

THANK YOU FOR YOUR TIME

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DATA ANALYST PRESENTATION